=== UNIT 2: CYTOSKELETON DYNAMICS, ORGANELLES & VESICULAR TRAFFICKING ===

>> 2.1 The Cytoskeletal Architecture: Three Distinct Filament Systems

The eukaryotic cytoplasm is structured by a dynamic polymeric filament network essential for mechanical integrity, cell division, organellar positioning, and intracellular trafficking.

* Microtubules: Hollow, cylindrical polymers of 25 nm outer diameter composed of 13 parallel protofilaments. Each protofilament is a linear polymer of alpha- and beta-tubulin heterodimers stacked head-to-tail, conferring structural polarity (plus end terminates with beta-tubulin; minus end terminates with alpha-tubulin). Microtubules originate from Microtubule Organizing Centers (MTOCs/Centrosomes) where gamma-tubulin ring complexes (gamma-TuRC) nucleate minus ends.
* Dynamic Instability: Microtubules alternate stochastically between slow growth (polymerization) and rapid catastrophe (depolymerization). Beta-tubulin binds GTP; when incorporated at the growing plus end, GTP is hydrolyzed to GDP. If polymerization rate exceeds hydrolysis, a stabilizing 'GTP Cap' is maintained. Loss of the GTP cap triggers conformational strain within the lattice, causing outward curling of protofilaments and rapid catastrophic disassembly. Rescue occurs when a new GTP cap re-forms.
* Microfilaments (Actin Filaments): Two-stranded helical polymers of globular actin (G-actin) with a diameter of 7 nm. Structurally polar with a barbed (plus) end and pointed (minus) end. Actin filaments undergo 'treadmilling' at steady state: G-actin-ATP monomers add preferentially to the barbed end (lower critical concentration, Cc = 0.12 microM) while G-actin-ADP dissociates from the pointed end (higher Cc = 0.6 microM). Regulated by actin-binding proteins: profilin (promotes ADP-to-ATP exchange), cofilin (severs ADP-actin filaments), and Arp2/3 complex (nucleates 70-degree branched actin networks).
* Intermediate Filaments: Non-polar, 10 nm ropelike fibers conferring mechanical tensile strength. Composed of cell-type specific fibrous proteins: keratins (epithelia), vimentin (mesenchymal cells), neurofilaments (neurons), and nuclear lamins (nucleoplasm). Subunits form antiparallel staggered tetramers, precluding directional polarity.

>> 2.2 Molecular Motor Proteins

* Kinesins: Plus-end-directed microtubule motors that mediate anterograde transport (from centrosome toward cell periphery). Move processively in discrete 8 nm steps utilizing ATP hydrolysis by two globular motor heads in a coordinated hand-over-hand mechanism.
* Dyneins: Minus-end-directed microtubule motors mediating retrograde transport (toward centrosome). Cytoplasmic dynein is a massive multisubunit complex (1.2 MDa) requiring the dynactin cofactor complex to bind cargo vesicles. Axonemal dynein

generates sliding forces in 9+2 cilia and flagella.
* Myosins: Actin-dependent molecular motors. Myosin II forms bipolar thick filaments in striated muscle; its S1 head hydrolyzes ATP to execute power strokes tilting the lever arm by 70 degrees, pulling actin filaments inward.

>> 2.3 The Endomembrane System and Protein Sorting

* Endoplasmic Reticulum (ER): Rough ER is studded with membrane-bound ribosomes synthesizing secretome, transmembrane, and lysosomal proteins. Nascent polypeptides bear an N-terminal hydrophobic Signal Peptide recognized by the Signal Recognition Particle (SRP). SRP arrests translation, binds the ER membrane SRP receptor, and transfers the ribosome-nascent chain complex to the Sec61 translocon channel for co-translational translocation into the ER lumen.
* Protein Folding and N-Glycosylation: Molecular chaperones (BiP/GRP78, calnexin, calreticulin) ensure proper folding. Protein disulfide isomerase (PDI) catalyzes disulfide bond formation. Oligosaccharyltransferase transfers a preformed 14-sugar oligosaccharide (Glc3Man9GlcNAc2) en bloc to recipient asparagine residues in the consensus sequence Asn-X-Ser/Thr. Misfolded proteins undergo ER-Associated Degradation (ERAD) via retrotranslocation to the cytosol and polyubiquitination for proteasomal destruction.
* Golgi Complex: Polarized organelle composed of cis, medial, and trans-Golgi cisternal stacks. Receives vesicles at the cis-Golgi network, modifies oligosaccharide chains (mannose trimming, addition of GlcNAc, galactose, sialic acid), and packages sorted cargoes at the trans-Golgi network (TGN).

>> 2.4 Vesicular Transport Machinery: Coats and SNAREs

* COPII-Coated Vesicles: Mediate anterograde transport from ER to ER-Golgi intermediate compartment (ERGIC) and cis-Golgi. Coat assembly is initiated by Sar1 GTPase, which recruits Sec23-Sec24 inner cargo-adaptor heterodimers followed by Sec13-Sec31 outer cage complexes.
* COPI-Coated Vesicles: Mediate retrograde retrieval transport from Golgi back to ER. Recruited by Arf1 GTPase. Recognizes C-terminal retrieval signals: KDEL on soluble luminal ER proteins (via KDEL receptor) and KKXX on transmembrane proteins.
* Clathrin-Coated Vesicles: Mediate transport from TGN to endosomes/lysosomes and from plasma membrane inward during endocytosis. Mannose-6-phosphate (M6P) tags on lysosomal enzymes are recognized by M6P receptors in TGN and incorporated into clathrin coats via GGA and AP1 adaptors.
* SNARE-Mediated Fusion: Vesicle targeting specificity is governed by Rab family small GTPases and tethering complexes. Fusion is executed by SNAREs: one v-SNARE (e.g., synaptobrevin) on vesicle pairs with three t-SNARE helices (e.g., syntaxin and SNAP-25) on target membrane to assemble a four-helix trans-SNARE bundle (coiled-coil). Bundle zippering overcomes electrostatic repulsion, forcing lipid bilayer hemifusion and complete fusion. Disassembly of the stable cis-SNARE complex

requires the AAA+ ATPase NSF and alpha-SNAP.

[EXAMPLE 1: Mechanism of Action of Anti-Mitotic Chemotherapeutics]

Problem Context: Taxol (paclitaxel) and Vinca alkaloids (vincristine, vinblastine) both disrupt the microtubule cytoskeleton to halt cancer cell proliferation, yet act through opposite biophysical mechanisms. Detail their molecular targets, cellular consequences, and therapeutic rationale.

Molecular Target 1 (Taxol): Paclitaxel binds stoichiometrically to the interior surface of the beta-tubulin subunit within the microtubule wall. It stabilizes lateral protofilament interactions, suppresses dynamic instability, and prevents microtubule depolymerization even in the absence of GTP. Mitotic spindles become frozen and cannot generate tension required to separate sister chromatids.

Molecular Target 2 (Vinca Alkaloids): Vincristine binds to tubulin heterodimers at the interface between alpha and beta subunits, inhibiting polymerization at high concentrations and inducing spiral aggregate formation. Microtubules depolymerize and mitotic spindles fail to assemble.

Cellular Outcome: In both cases, kinetochores remain unattached or lack tension. The Mad2/BubR1 spindle assembly checkpoint (SAC) remains persistently active, sequestering Cdc20 and inhibiting the Anaphase-Promoting Complex/Cyclosome (APC/C). Cells arrest in prometaphase and activate mitochondria-dependent intrinsic apoptosis.

Conclusion: Manipulating microtubule catastrophe/assembly kinetics selectively eradicates rapidly dividing neoplasias by triggering SAC-mediated mitotic catastrophe.

[EXAMPLE 2: I-Cell Disease (Inclusion Cell Disease / Mucolipidosis II)]

Problem Context: Describe the molecular etiology, biochemical defect, and cellular phenotype of I-Cell Disease, an autosomal recessive lysosomal storage disorder.

Normal Targeting Pathway: In the cis-Golgi, lysosomal hydrolases are recognized by UDP-N-acetylglucosamine:lysosomal-enzyme N-acetylglucosamine-1-phosphotransferase (GlcNAc phosphotransferase). It recognizes a specific 3D protein conformation epitope (signal patch) and transfers GlcNAc-1-phosphate to terminal mannoses of N-linked glycans. In the trans-Golgi, phosphodiesterase removes GlcNAc, leaving exposed Mannose-6-Phosphate (M6P) residues that bind M6P receptors for clathrin-mediated lysosomal packaging.

Biochemical Lesion in I-Cell Disease: Loss-of-function mutation in the GNPTAB gene encoding the alpha/beta subunits of GlcNAc phosphotransferase. Lysosomal enzymes fail to acquire the M6P targeting signal.

Pathological Outcome: Unmodified acid hydrolases are not sorted into transport vesicles at the TGN. Instead, they are routed through the default constitutive secretory pathway and dumped into the extracellular bloodstream.

Phenotype: Lysosomes lack essential degradation enzymes and become engorged with

undigested glycosaminoglycans, lipids, and proteins, forming dense intracellular inclusion bodies ('I-cells'). Patients develop severe skeletal deformities (dysostosis multiplex), coarse facial features, hepatosplenomegaly, and fatal cardiopulmonary failure in early childhood.

[EXAMPLE 3: Botulinum and Tetanus Neurotoxins as Molecular Probes of SNAREs]

Problem Context: Botulinum neurotoxin (BoNT) and Tetanus neurotoxin (TeNT) are the most lethal bacterial toxins known, produced by Clostridium botulinum and Clostridium tetani. Detail how their enzymatic cleavage of SNARE proteins impairs neurotransmission.
Toxin Structure: Both toxins are 150 kDa zinc endopeptidases comprising a 100 kDa Heavy Chain (mediates cell binding and translocation) and a 50 kDa Light Chain (catalytic metalloprotease possessing a HEXXH zinc-binding motif).
Cleavage Targets:
- BoNT/A and BoNT/E: Cleave SNAP-25 (a t-SNARE on the presynaptic plasma membrane) at specific C-terminal sites, destroying its ability to form stable 4-helix bundles with syntaxin and synaptobrevin.
- BoNT/B, BoNT/D, BoNT/F, BoNT/G, and TeNT: Cleave Synaptobrevin / VAMP-2 (the v-SNARE on synaptic vesicles).
- BoNT/C: Cleaves Syntaxin-1 and SNAP-25.
Physiological Consequences:
- Botulism: BoNT acts locally at peripheral neuromuscular junctions. Inability of acetylcholine vesicles to fuse with the presynaptic terminal eliminates motor endplate potentials, producing flaccid paralysis and respiratory failure.
- Tetanus: TeNT undergoes retrograde axonal transport along motor neurons into spinal cord inhibitory interneurons (Renshaw cells). It cleaves VAMP-2, abolishing exocytosis of inhibitory neurotransmitters (GABA and glycine). Motor neurons fire uncontrollably, producing spastic paralysis and tetanic muscle spasms.
Conclusion: Intact four-helix trans-SNARE bundles are an absolute thermodynamic prerequisite for biological membrane fusion.

[EXAMPLE 4: Energetics and Processivity of Kinesin-1 Motor Stepping]

Problem Context: Conventional Kinesin-1 carries membrane organelles along microtubules. Using single-molecule optical trap data, explain how kinesin converts ATP free energy into mechanical work over an 8.2 nm step against a 6 pN load. Calculate its thermodynamic efficiency.
Biophysical Steps:
1. Microtubule binding: Leading head is bound to beta-tubulin without nucleotide; trailing head holds ADP.
2. ATP Binding: ATP binds leading head, inducing conformational docking of the 15-amino-acid neck linker along the catalytic core toward the plus-end.

3. Power Step: Neck linker docking throws the trailing head forward by 16 nm (moving the center of mass forward by 8.2 nm, the distance between adjacent tubulin heterodimers).
4. Hydrolysis and Release: The new leading head binds tubulin, releases ADP, while the new trailing head hydrolyzes ATP to ADP+Pi and detaches.
Thermodynamic Work Calculation:
Mechanical Work (W) = Force * Distance = F * d
W = (6 * 10^-12 N) * (8.2 * 10^-9 m) = 4.92 * 10^-20 Joules per step.
Available Chemical Free Energy from ATP Hydrolysis:
delta G_ATP under cellular conditions = -50 kJ/mol = (50,000 J) / (6.022 * 10^23) = 8.30 * 10^-20 Joules per molecule.
Thermodynamic Efficiency (eta):
eta = (Mechanical Work Done / Chemical Energy Consumed) * 100
eta = (4.92 * 10^-20 J / 8.30 * 10^-20 J) * 100 = 59.3%.
Conclusion: Kinesin operates at ~60% thermodynamic efficiency, vastly surpassing human-engineered combustion engines (~25-30%), with strict 1:1 coupling between ATP hydrolyzed and 8.2 nm steps taken.

>> 2.5 University Review & Examination Questions

* Q1: Differentiate between COPI, COPII, and Clathrin vesicular transport pathways regarding their coat proteins, GTPases, direction of movement, and biological cargo.
Answer: COPII utilizes Sar1 GTPase and Sec23/Sec24 + Sec13/Sec31 coats for anterograde transport from ER to cis-Golgi carrying newly synthesized proteins. COPI utilizes Arf1 GTPase and coatomer complexes for retrograde retrieval transport from Golgi to ER carrying escaped ER-resident proteins with KDEL/KKXX motifs. Clathrin utilizes Arf1 or dynamin and AP1/AP2/GGA adaptors for bidirectional trafficking between TGN, endosomes, and plasma membrane, transporting lysosomal enzymes (M6P-tagged) and endocytosed receptors.
* Q2: Explain the molecular mechanism of actin filament treadmilling and how it generates protrusion forces at the leading edge of migrating cells.
Answer: Actin treadmilling occurs when the free G-actin monomer concentration is intermediate between the critical concentrations of the barbed plus end (Cc+ = 0.12 microM) and pointed minus end (Cc- = 0.60 microM). Subunits add continuously at the plus end while dissociating from the minus end at an equal rate, maintaining constant filament length. In migrating cells, the Arp2/3 complex binds to pre-existing actin filaments upon activation by WASp/WAVE and nucleates new filaments at a 70-degree angle. Rapid plus-end polymerization against the plasma membrane generates mechanical pushing forces that project the lamellipodium forward.

>> 2.6 Key Takeaways and Revision Points

* Microtubules (25 nm, tubulin heterodimers) display dynamic instability driven by the GTP cap at their plus end.
* Actin filaments (7 nm) treadmill and coordinate cell motility via Arp2/3, profilin,

and cofilin.

* Kinesin moves cargo toward plus ends (anterograde); Dynein moves cargo toward minus ends (retrograde).
* SNARE complexes (1 v-SNARE + 3 t-SNARE coils) drive membrane fusion, disassembled post-fusion by NSF/alpha-SNAP.